Anurag Nepal

nepalanurag72@gmail.com | (650) 270-6636 | San Francisco, CA
github.com/nepalanurag | linkedin.com/in/anurag-nepal-1587a4191 | anurag-nepal-portfolio.vercel.app

PROFESSIONAL SUMMARY

ML engineer and data scientist, M.S. Statistical Data Science (4.0 GPA, Distinguished Graduate). 1.5 years building ML models for cancer genomics and medical imaging, including a peer-reviewed CNN publication. Strong in Python, PyTorch, statistical modeling, reproducible research.

TECHNICAL SKILLS

Python (PyTorch, TensorFlow, scikit-learn, Keras, OpenCV), R, SQL, Java, C/C++. Penalized regression, survival analysis, hypothesis testing, CNNs, classification, forecasting, cross-validation. Git, Jupyter, BigQuery, experiment tracking.

EXPERIENCE

Graduate Assistant, Dept. of Mathematics, San Francisco State University Feb 2025 - May 2026

- Permutation-assisted group lasso for multi-omics feature selection on TCGA-BRCA: FDR 0.01 across 18 scenarios vs. 0.53 for lasso; survival models reached 0.948 AUC; presented at BBSW 2025

SELECTED PROJECTS

Thesis Findings: Multi-modal Feature Selection Live Demo

- M.S. thesis on permutation-assisted group lasso, validated on TCGA breast cancer data

Biomedical Imaging Analysis Live Demo

- COVID-19 lung infection segmentation and quantification on chest CT, in-browser demo

Malaria Detection using ML Live Demo

- CNN classifier for malaria parasites in blood smear images (peer-reviewed publication)

Kalshi Backtest Lab Live Demo

- Statistically rigorous backtesting of Kalshi BTC strategies (130M trades, BigQuery)

ECG Classification Live Demo

- ECG heartbeat classification with CNN, ANN, KNN on MIT-BIH, in-browser demo

Lasso Simulation Study Live Demo

- Simulation study of lasso variable selection across sample sizes and dimensions

Goal Difference Live Demo

- Football predictions for Europe top 5 leagues (Poisson, XGBoost)

AI Lab Live Demo

- Interactive dashboards for statistics-meets-AI experiments

EDUCATION

M.S. Statistical Data Science, SFSU, 2024-2026, GPA 4.0/4.0, Distinguished Graduate Award
B.E. Computer Science and Engineering, NMIT, 2019-2023, GPA 9.34/10.0, top 5 graduates

CERTIFICATIONS

Data Analyst (Datacamp), Intermediate SQL (Datacamp), Python (Coursera)

PUBLICATION

- Co-author, "Detection of Malaria in Blood Cells using CNN" (peer-reviewed)